A LOCALLY FINITELY PRESENTED COHEN-MACAULAY MORPHISM SPLITS INTO OPEN AND CLOSED PIECES OF CONSTANT RELATIVE DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6435-6443 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6445-6468, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Cohen-Macaulay morphisms of algebraic spaces, relative dimension, and open and closed decompositions; the construction is retained.

Source reference. Original lemma 02NM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6468–6468. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume that f is locally of finite presentation and Cohen-Macaulay. Then there exist open and closed subschemes $X_d \subset X$ such that $X = \coprod_{d \geq 0} X_d$ and $f|_{X_d} : X_d \to Y$ has relative dimension d.*

Proof. Choose a commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

with étale vertical arrows and U and V schemes. Then $U \to V$ is locally of finite presentation and Cohen-Macaulay (immediate from our definitions). Thus we have a decomposition $U = \coprod_{d \geq 0} U_d$ into open and closed subschemes with $f|_{U_d} : U_d \to V$ of relative dimension d, see Morphisms, Lemma 02NM. Let $u \in U$ with image $x \in |X|$. Then f has relative dimension d at x if and only if $U \to V$ has relative dimension d at u (this follows from our definitions). In this way we see that U_d is the inverse image of a subset $X_d \subset |X|$ which is necessarily open and closed. Denoting X_d the corresponding open and closed algebraic subspace of X we see that the lemma is true. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.